



THE CITIZENS FOUNDATION

• Alumni Pathways • Tertiary Support Program

Career Guidance Book



HOW TO USE THIS GUIDE

This Career Guidance Book is your starting point; not your finishing point. It is designed to give you a clear overview of the fields available to you, the programs you can study, and the kinds of careers they lead to. But the research does not stop here.

STEP 1 — Start here

Read through the section for each field that interests you. Look at the programs, the road map, and the career paths. Ask yourself: does this feel like something I could see myself doing?

STEP 2 — Go deeper with university websites

Once you have a shortlist of fields, visit the official websites of the universities listed under 'Universities on Policy'. Read about their specific programs, admission requirements, fee structures, and campus life.

STEP 3 — Use Google and AI tools

Search for things like 'what does a [career name] do in Pakistan' or 'difference between CS and Software Engineering'. Use AI tools like Claude or Google to ask follow-up questions and explore further. These tools can help explain concepts in simple language.

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- Electrical & Electronics Engineering
- Chemical & Petrochemical Engineering
- Mechanical Engineering
- Civil & Environmental Engineering
- Marine Engineering
- Industrial Engineering
- Aerospace & Aeronautical Engineering
- Materials, Mining & Textile Engineering

2. COMPUTER STUDIES

- Computer Science
- Information Technology
- Computer Engineering
- Software Engineering

3. MANAGEMENT SCIENCES

- Business Administration & Management
- Finance & Economics
- Tourism & Hospitality

4. SOCIAL SCIENCES

- Architecture
- Art & Design
- Education
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- Health & Physical Education
- Political & Social Sciences
- Islamic & Comparative Religious Studies
- Language & Literature
- Law & Criminal Studies
- Media Sciences & Communication

5. HEALTH SCIENCES

- Medicine
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- Allied Health
- Pharmaceutical & Nutritional Sciences
- Vision Care, Diagnostic & Technical

6. NATURAL SCIENCES

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- Statistics & Mathematics
- Chemistry, Physics & Marine Sciences

7. AGRICULTURAL SCIENCES

ENGINEERING

BRIEF DESCRIPTION

Engineering is the application of mathematics, science, and technology to design, build, and improve systems, structures, machines, and processes that solve real-world problems. Engineers work across industries including infrastructure, energy, manufacturing, technology, aviation, textiles, and chemicals.

Program Duration: 4 years (8 semesters). Requires strong problem-solving skills, consistency, and interest in technical subjects.

ROAD MAP

Matric (Science) Focus: Math, Physics, Chemistry	FSc Pre- Engineering Intermediate (2 years)	Entry Test ECAT / NTS / University test	BE / BS Engineering 4 years	Career / MS Industry or further study
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PROGRAMS

Core Engineering Disciplines:

- Civil & Environmental Engineering
- Electrical & Electronics Engineering
- Mechanical Engineering
- Chemical & Petrochemical Engineering
- Industrial Engineering
- Computer Engineering

Specialized Engineering Fields:

- Marine Engineering
- Aerospace & Aeronautical Engineering
- Materials Engineering
- Mining Engineering
- Textile Engineering

CAREER PATHS

Civil Engineer	Electrical Engineer	Mechanical Engineer	Project Manager	Research & Development	Environmental Engineer
Chemical Engineer	Software Engineer	Industrial Engineer	Quality Control Officer	Construction Manager	Energy Consultant

COMPUTER SCIENCE

BRIEF DESCRIPTION

Computer Studies focuses on the use of computers, software, data, and networks to solve problems, build digital solutions, and manage information. This field is ideal for students with an interest in technology, logic, problem-solving, and innovation.

Program Duration: 4 years (8 semesters). Emphasises practical skills, programming, and continuous learning.

ROAD MAP

Matric (Science) Focus: Math, Computer Science	FSc Pre-Engineering / ICS Intermediate (2 years)	Entry Test NTS / University-specific test	BS CS / SE / IT 4 years	Career / MS Tech industry or further study
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PROGRAMS

Core Programs

- Computer Science (CS)
- Software Engineering (SE)
- Information Technology (IT)
- Computer Engineering

Emerging & Specialised Fields

- Artificial Intelligence & Machine Learning
- Cyber Security
- Data Science & Analytics
- Game Development & Graphics
- Business & Data Analytics

CAREER PATHS

Software Developer	Web / App Developer	Data Analyst	UX/UI Designer	Network Engineer	Database Administrator
Cybersecurity Analyst	IT Manager	AI/ML Engineer	DevOps Engineer	Product Manager (Tech)	Cloud Solutions Architect

MANAGEMENT SCIENCES

BRIEF DESCRIPTION

Management Sciences equip students with skills in strategic thinking, leadership, problem-solving, data analysis, and decision-making. Graduates learn how organisations operate, how resources are managed, and how business challenges are solved across sectors including finance, marketing, HR, consulting, and entrepreneurship.

Program Duration: 4 years. Combines theory, case studies, presentations, and practical projects.

ROAD MAP

Matric Any group (Science or Arts)	FSc / FA / ICS / ICom Intermediate (2 years)	Entry Test SAT / University-specific test	BBA / BS Economics / BS Finance 4 years	Career / MBA / MS Industry or postgraduate
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PROGRAMS

Core Programs

- Bachelor of Business Administration (BBA)
- BS Accounting & Finance
- BS Economics
- BS Economics & Mathematics
- BS Digital Marketing

Emerging & Specialised Fields

- Supply Chain Management
- Human Resource Management
- Marketing & Digital Marketing
- Entrepreneurship
- Finance & FinTech
- Data & Business Analytics

CAREER PATHS

Finance Executive	Management Trainee Officer (MTO)	Operations Trainee	HR Assistant	Digital Marketing Executive	Research Analyst
Junior Accountant	Business Analyst	Procurement Assistant	Talent Acquisition Officer	Business Development Executive	Sales Executive

SOCIAL SCIENCES

BRIEF DESCRIPTION

Social Sciences focus on understanding people, society, culture, law, communication, governance, and education. These disciplines develop critical thinking, research, communication, and social awareness, preparing graduates for careers in policy, media, education, law, development work, and creative industries.

Program Duration: 4 years. Emphasises reading, writing, discussion, research, and fieldwork.

ROAD MAP

Matric Science or Arts group	FA / FSc / ICS Intermediate (2 years)	Entry Test Varies by university & program	BS / LLB / BFA / B.Ed 4 years (LLB = 5 years)	Career / MS / MPhil Public sector, NGOs, media, law
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PROGRAMS

Core Programs

- Education
- Law & Criminal Studies
- Political Science
- International Relations
- Sociology
- Psychology
- Social Work

Emerging & Specialised Fields

- Media Sciences & Mass Communication
- Journalism
- Language & Literature (English, Urdu, Linguistics)
- Art & Design
- Architecture
- Fashion & Textile Design

CAREER PATHS

Lawyer / Legal Advisor	Journalist / Reporter	Counsellor	Graphic Designer	Creative Writer	Research Analyst
Teacher / Educator	Policy Analyst	Social Worker	Fashion Designer	Media Assistant	Training Coordinator

MEDICAL SCIENCES

BRIEF DESCRIPTION

Health Sciences focus on preventing, diagnosing, treating, and managing illnesses, as well as supporting patient care, rehabilitation, diagnostics, and public health. These fields combine science, clinical training, and service, and require strong discipline, empathy, and commitment.

Program Duration: 4–5 years depending on field. Some programs require licensing exams, internships, or house jobs.

ROAD MAP

Matric (Science) Focus: Biology, Chemistry, Physics	FSc Pre-Medical Intermediate (2 years)	Entry Test MDCAT / NTS / University test	MBBS / BDS / Allied Health 4–5 years	House Job / Licensing / MS Clinical practice or specialisation
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PROGRAMS

Clinical & Core Medical Fields

- Medicine (MBBS)
- Dentistry (BDS)
- Nursing (BSN)
- Doctor of Physical Therapy (DPT)
- Doctor of Pharmacy (Pharm-D)

Allied Health & Diagnostic Sciences

- Medical Laboratory Technology (MLT)
- Radiological Sciences & Imaging
- Optometry
- Speech & Language Pathology
- Dental Technology & Hygiene
- Biotechnology
- Occupational Therapy

CAREER PATHS

Medical Doctor (MBBS)	Nurse / Nursing Officer	Pharmacist	Lab Technologist	Medical Technician	Public Health Officer
Dentist	Radiologist	Physio-therapist	Biotech Researcher	Clinical Coordinator	Hospital Administrator

NATURAL SCIENCES

BRIEF DESCRIPTION

Natural Sciences study the laws of nature and living systems, helping us understand how the physical and biological world works. These disciplines build strong foundations in scientific thinking, research, data analysis, and laboratory skills, and often serve as pathways to research, teaching, healthcare, industry, and postgraduate studies.

Program Duration: 4 years. Requires strong interest in science, experimentation, and analysis.

ROAD MAP

Matric (Science) Focus: Biology, Chemistry, Physics, Math	FSc Pre-Medical or Pre-Engineering Intermediate (2 years)	Entry Test NTS / University test	BS Natural Sciences 4 years	MS / MPhil / PhD or Career Research, academia, or industry
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PROGRAMS

Physical & Quantitative Sciences - Physics - Chemistry - Statistics - Mathematics	Biological & Life Sciences - Microbiology - Zoology - Botany - Marine Sciences
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CAREER PATHS

Research Scientist	Mathematician	Statistician
Quality Assurance Officer	Science Teacher / Lecturer	Lab Technician

FREE LEARNING RESOURCES

Below are free online resources organised by discipline. Use these alongside your university research to build subject knowledge, explore career paths, and strengthen your preparation.

I want to explore what different careers actually look like

Not sure what engineers, doctors, or designers actually do every day? Start here.

CrashCourse ([YouTube](#)) – Short, engaging videos on almost every field – Business, Psychology, Engineering, Medicine, and more

TED-Ed ([YouTube](#) / [ted.com](#)) – Real professionals talk about their work and ideas in plain language

LinkedIn Learning previews ([LinkedIn app](#)) – Search any job title and watch free preview videos of what that career involves

I want to start learning the basics of my field

Once you have a field in mind, these platforms let you take real university-level courses for free.

Khan Academy ([khanacademy.org](#)) – Covers Math, Science, Computing, Economics – everything explained from scratch, at your own pace

Coursera ([coursera.org](#)) – Choose "Audit" on any course to access all videos and readings for free. Covers every major field

edX ([edx.org](#)) – Free courses from MIT, Harvard, and other universities – audit mode is always free

I want to improve my English

Strong English will help you in university, in interviews, and in your career. These make it easy to practise daily.

BBC Learning English ([bbc.co.uk/learningenglish](#)) – Short daily lessons, news-based listening practice, and grammar exercises

Duolingo ([app](#)) – A few minutes a day builds vocabulary and reading confidence over time

EnglishClass101 ([YouTube](#)) – Clear lessons on grammar, vocabulary, and spoken English for different levels

I want to prepare for entry tests and admissions

Entry tests like ECAT, MDCAT, and BCAT require consistent practice.

PakMCQs ([pakmcqs.com](#)) – Pakistan-specific MDCAT, ECAT, and NTS past papers and practice questions

Khan Academy Math & Science ([khanacademy.org](#)) – Revise the topics that come up in entry tests – Algebra, Physics, Chemistry, Biology

YouTube – search your test name ([YouTube](#)) – Search "MDCAT preparation 2025" or "ECAT tips" to find Pakistani students and teachers sharing strategies